5.23 Tyre rotation

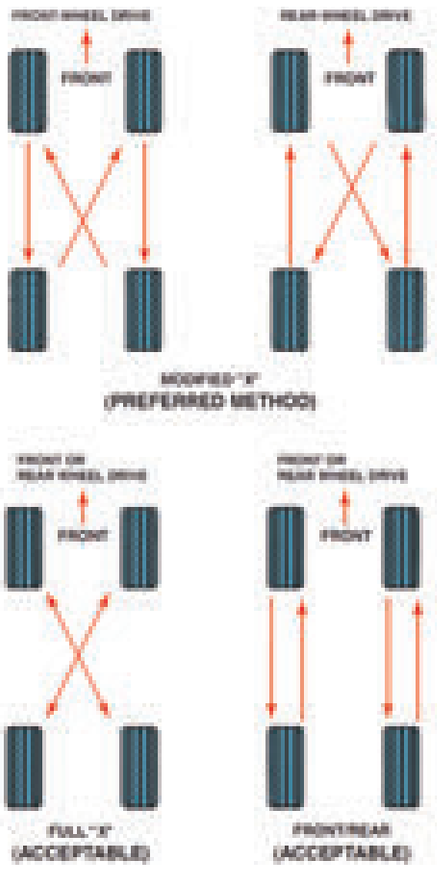


Figure 5.23 Wheel Rotation procedure

In general the tyres in the vehicle are subjected to wear but the amount of wear on the four wheels is not uniform. The tyres in the rear wheels of the vehicle wear more quickly than the front wheels due to the power transmission to the rear axle. Right side wheels generally wear twice in amount than the left wheels. Hence, based on the manufacturers' guidelines tyres are rotated and fitted in the vehicles after the vehicle is travelled to certain kilometers. In cars and commercial vehicles for about 8000 km of the vehicle's travel the tyre is changed based on rotational mode.

The right side of the rear wheel tyre is changed to left front tyre. Left side front tyre is changed to right front tyre. The right front wheel is kept as a spare wheel. The other wheel is changed to right rear wheels. This method of changing the wheels is called as tyre rotation. The figure 5.19 indicates the method of tyre rotation.

5.24 Wheel balancing

If the wheels are allowed to rotate then must rotate uniformly and smoothly without any vibrations. If there is no proper rotation of the wheels the weight of the rim must be balanced for uniform rotation of the wheels and this is method is called as wheel balancing. The rim weight is adjusted by moving different lead metal weights on the rim. Figure 5.24 shows the balance weight placed on the rim



Figure 5.24 Wheel balancing

Advantages of wheel balancing

1. It reduces the tyre wear.
2. It helps in steering the vehicle easily.
3. It reduces wobbling of the vehicle.

Types

1. Static balancing
2. Dynamic balancing

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